

Mathematical Modelling for Sustainable Development Healthcare Allocation

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1 Introduction

The trust of a hospital in mental health services struggles with operational issues caused by unpredictable walk-in patients and scheduled appointments, leading to bottlenecks and long waiting times. As a result, waiting times have increased and inefficiencies significantly impact outpatient services. The hospital is now interested in exploring how mathematical modeling can optimize resource allocation to improve service delivery. This report will present a mathematical modelling approach to address the problem and improve patient flow efficiently in the hospital to help optimize their resources.

2 Problem Statement

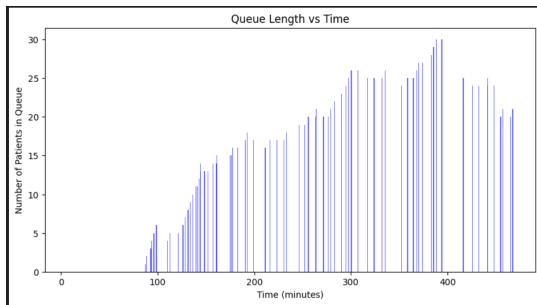
Currently, the clinic has four clinicians who perform 30-minute appointments. Patients can arrive through pre-booked appointments or walk-ins, with the rates varying from 2 to 12 patients per hour. Management aims to balance demand with capacity while ensuring timely care without overburdening staff.

3 Methodology

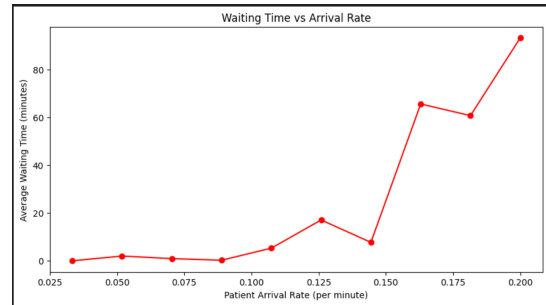
We applied the queueing model to model the clinic queue. M/M/c is a useful queueing model for multi-server queues. We analyze key performance metrics:

- **Utilization Factor (ρ):** Measures the workload of clinicians relative to their capacity.
- **Waiting Time (W):** Estimates the expected time a patient spends waiting before receiving care.
- **Probability of Waiting (P):** Determines the likelihood of patients experiencing delays.

4 Finding



(a) Queue vs Time



(b) Waiting time vs Arrival Rate

Figure 1: Graph showing the queue length at particular time and waiting period of a patient based on arrival rate

After running a queueing model in Python, we found that the clinic operates with 4 clinicians and serves on a first come first serve basis. The model showed that fluctuations in the arrival rates from 2 patients to 12 patients per hour lead to long waiting periods, shown in Figure 1b. The waiting period for a patient can go more than 1 hour with an arrival rate of 12 patients per hour. Single queues can make it difficult to operate effectively. Figure 1a illustrates the number of patients in the queue waiting for service for a given instance of time.

5 How we can Improve

To improve the service and optimize resources, two possible strategies can be applied, splitting the resources or dynamic allocation.

- **Splitting the resources:** Assuming that half of the patients have scheduled appointments and half are walk-ins, we allocate two clinicians for walk-ins and two for appointments.
- **Dynamic allocation:** The number of clinicians assigned to each group can adapt dynamically to meet varying demand levels. This flexibility ensures that we provide the best possible care when and where it is needed most.

6 Strategy Implementation

After implementation of two strategies using Python to simulate, we obtain the following graph, Figure 2. The graph illustrates various queueing strategies.

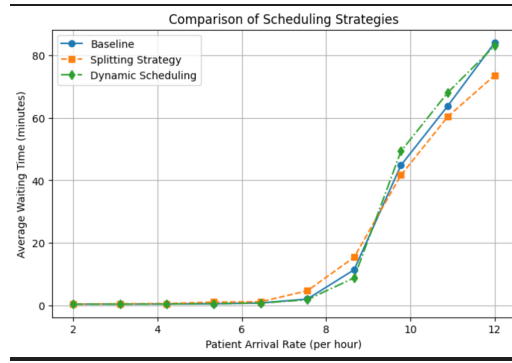


Figure 2: Comparison of Scheduling Strategies

- **Baseline:** The blue line represents the baseline scenario, where the waiting time increases as patient arrival rates increase.
- **Splitting:** The orange line represents the splitting strategy, which reduces waiting times, but is only effective for appointments scheduled patients.
- **Dynamic:** For walk-in patients, servers may be underutilized when demand is low. The green line represents the dynamic queueing strategy, which performs best during periods of high patient arrival rates.

7 Conclusion

The hospital trust was experiencing operational challenges stemming from an increase in patient numbers and limited capacity for expansion. This report utilized mathematical modelling to analyze the situation and propose potential solutions to help hospital management optimize resources. The M/M/c queueing model indicated that as patient arrival rates increased from 2 to 12 per hour, the waiting period exceeded 1 hour. To address this, it was proposed to split the 4 clinicians into 2 for walk-in patients and 2 for appointment-based patients. Another solution was the dynamic allocation of clinicians, which resulted in lower waiting times at higher arrival rates, making it more effective than other strategies.